

# Ethanol Blending in Gasoline – Nigeria

October 2025



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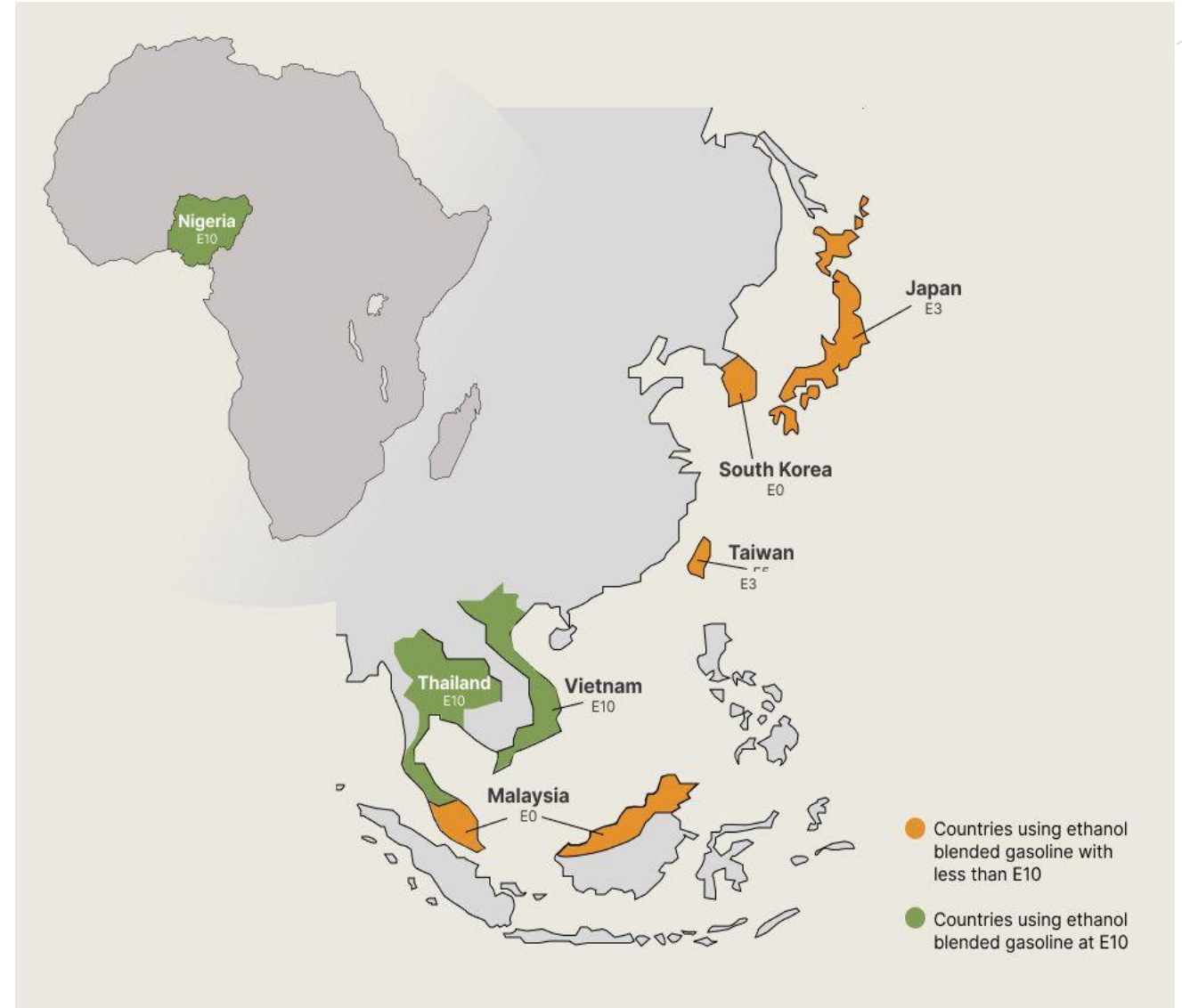


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# Ethanol Blending in Asia-Africa

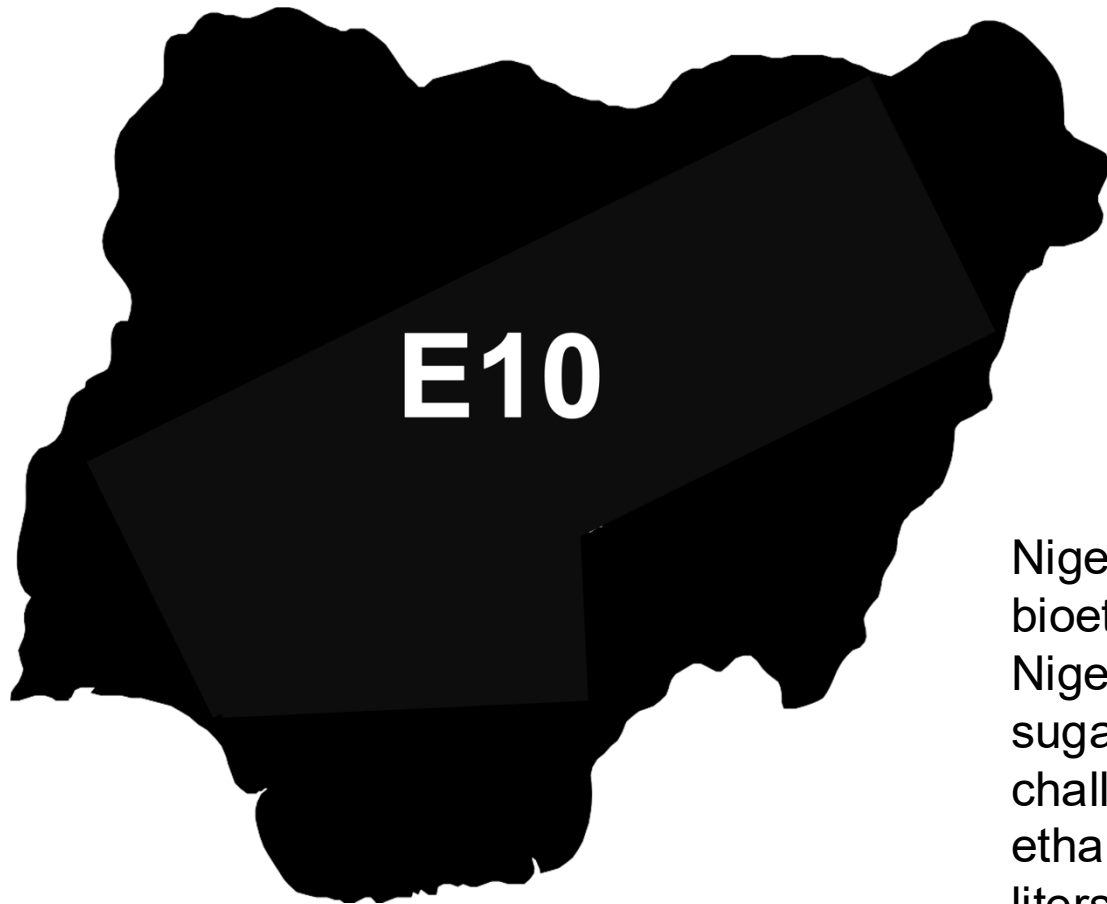
There are important fuel quality and environmental impact of vehicle emission challenges in these regions.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Asia and Africa to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Seven countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

# Ethanol Gasoline Blending in Nigeria

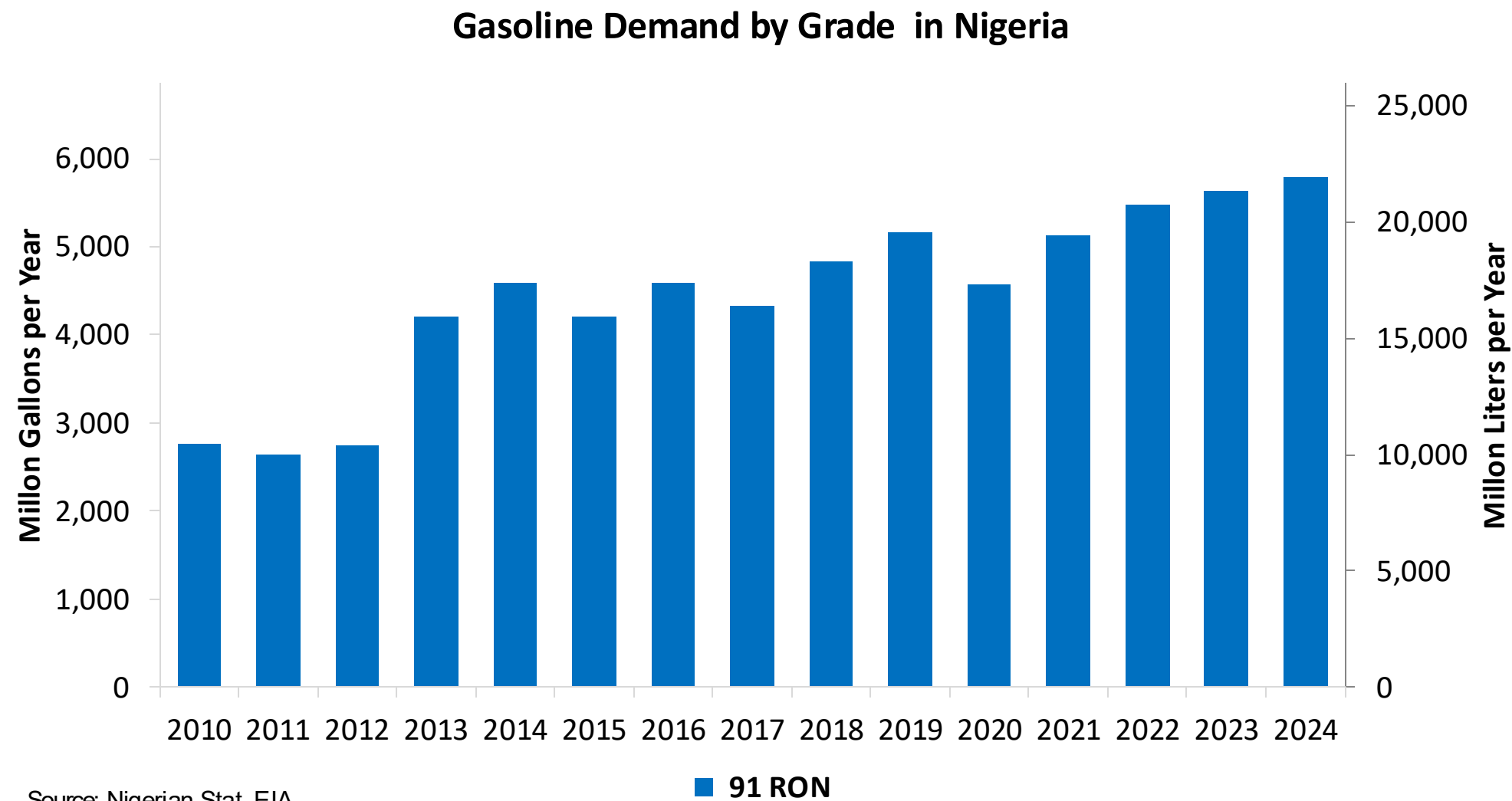


In 2024, gasoline consumption in Nigeria was 21,894 million liters (5,784 million gallons) and growth rate was 2.3%. Gasoline production and net imports were 9,242 and 12,652 million liters (2,441 and 3,342 million gallons) correspondingly. Nigeria offers one main grade: RON 91. Production and supply of a 95 RON grade will be commercially available in the short term.

Nigeria has a Biofuels Policy that aims for a 10% blend of bioethanol in gasoline, but this target has not been met. Nigeria produces bioethanol primarily from cassava and sugarcane, but the industry is small-scale and faces challenges, resulting in the country being a net importer of ethanol. Ethanol consumption in Nigeria was 379 million liters (100 million gallons) representing 1.7% of gasoline demand.

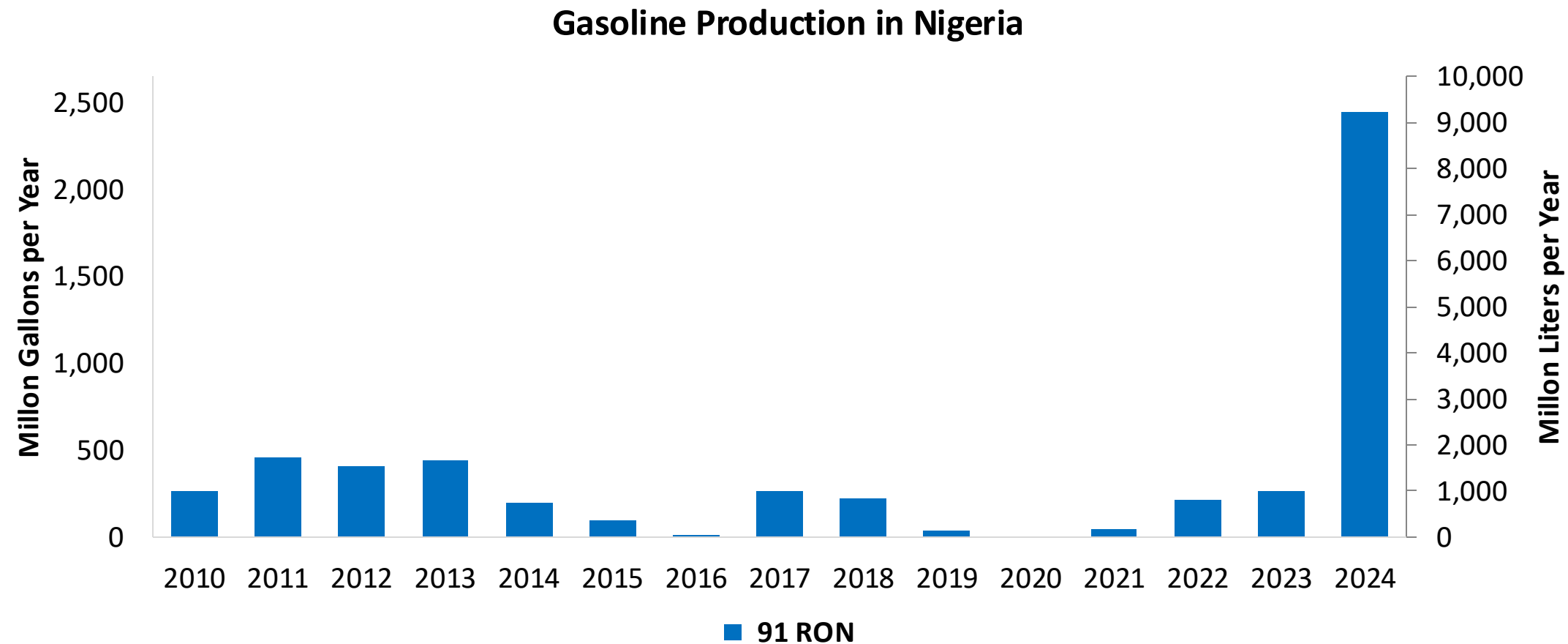
Source: Nigerian Stat, EIA, Climate Laws, Advanced Biofuels

# Gasoline Demand in Nigeria



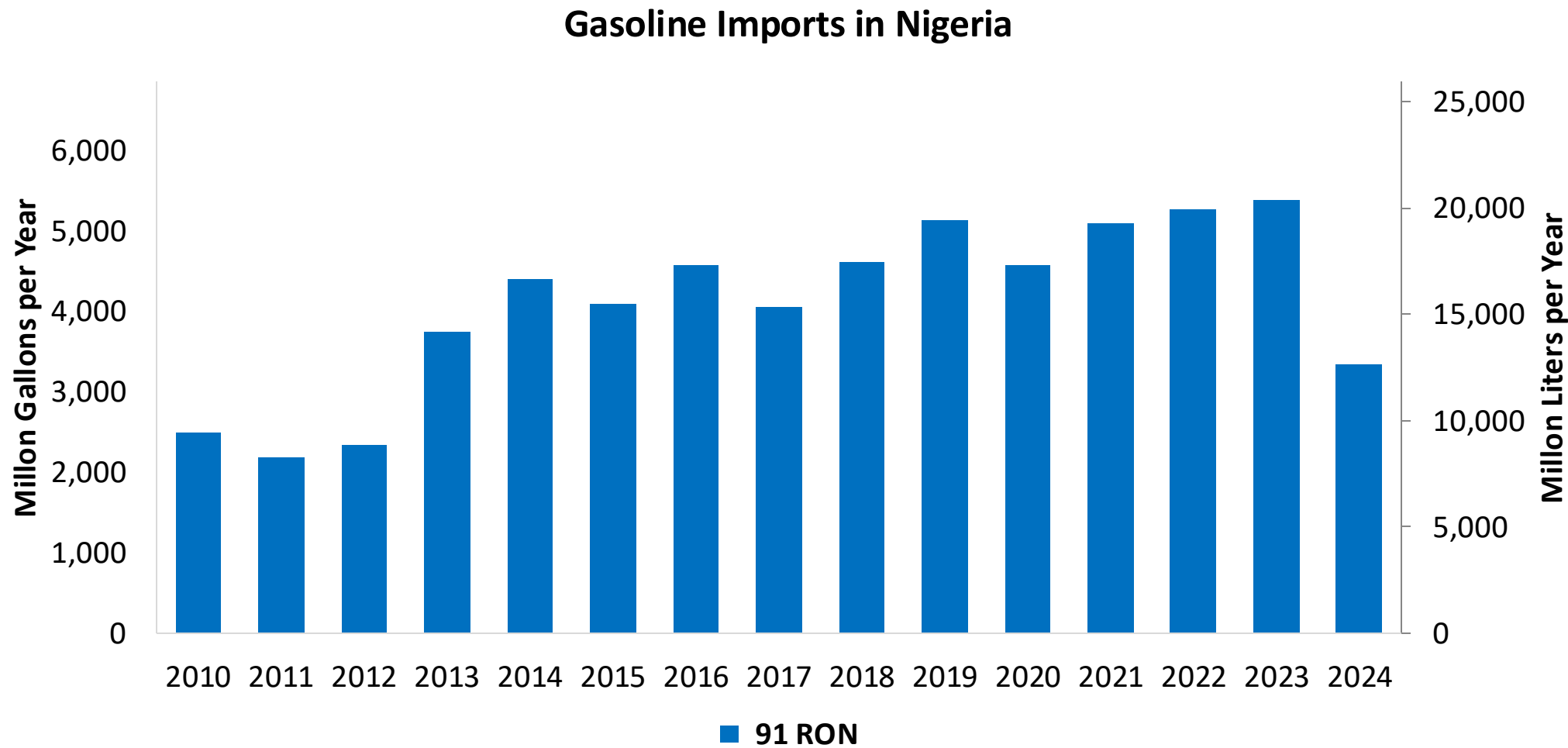
Source: Nigerian Stat, EIA

# Gasoline Production in Nigeria



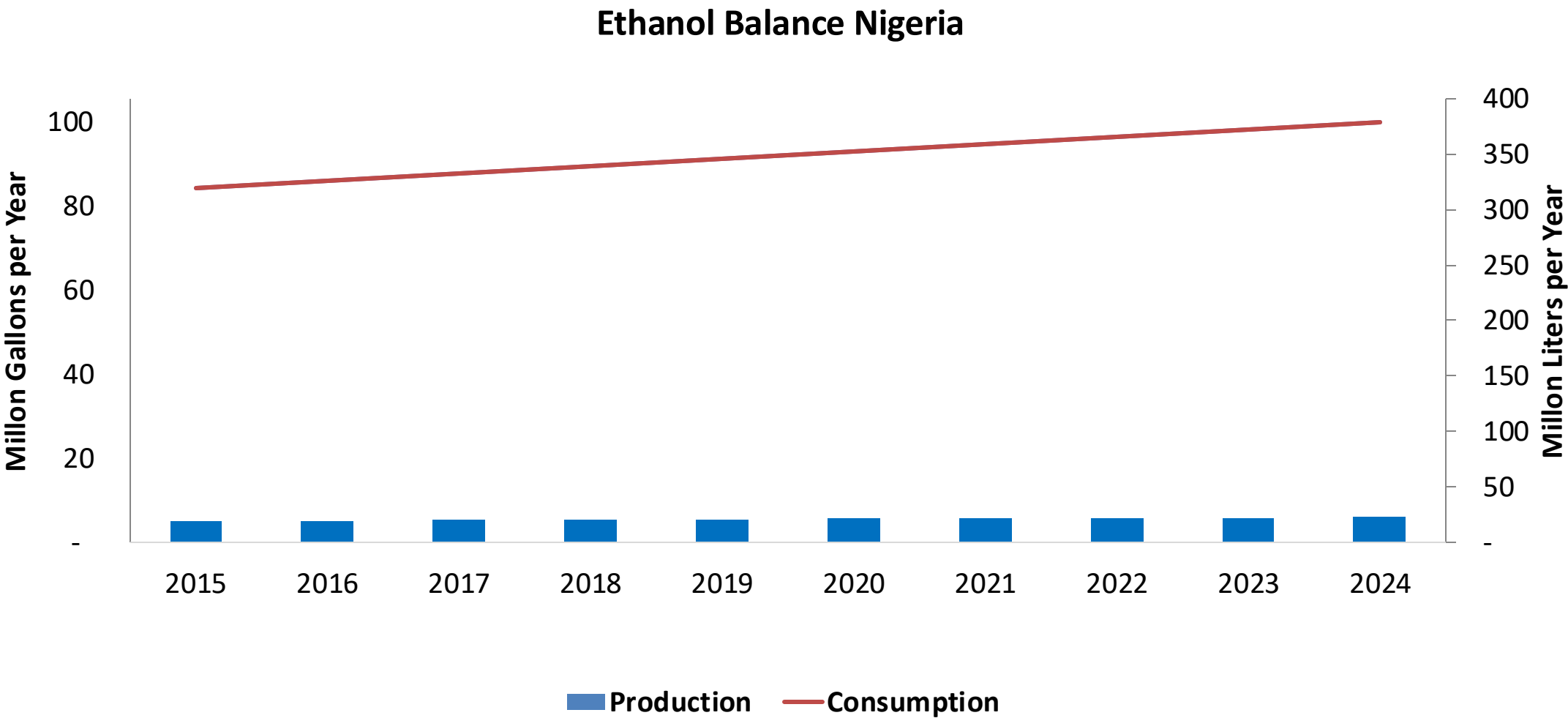
Source: Nigerian Stat, EIA

# Gasoline Imports in Nigeria



Source: Nigerian Stat, EIA

# Ethanol Balance in Nigeria



Source: Nigerian Stat, Climate Laws, Advanced Biofuels

# Gasoline Quality in Nigeria

| Name                | NIS116:2017. E10 (NIS 534:2006) |       |         |       |
|---------------------|---------------------------------|-------|---------|-------|
| Implementation Year | 2017                            |       |         |       |
| Applicability       | Nationwide                      |       |         |       |
| Grade               | PMS                             |       | PMS E10 |       |
|                     | Min                             | Max   | Min     | Max   |
| RON                 | 91.0                            |       | 92.0    |       |
| MON                 | 81.0                            |       | 82.0    |       |
| AKI                 |                                 |       |         |       |
| Benzene (%vol)      |                                 | 2.0   |         | 2.0   |
| Aromatics (% vol)   |                                 |       |         |       |
| Olefins (%vol)      |                                 |       |         |       |
| Lead (g/L)          |                                 | 0.002 |         | 0.002 |
| Manganese (mg/L)    |                                 | -     |         | -     |
| Sulfur (ppm)        |                                 | 150   |         | 150   |
| Oxygen (%v)         |                                 | 0.2   |         | 3.5   |
| Ethanol (%vol)      |                                 |       |         | 10.0  |
| MTBE (%vol)         |                                 |       |         |       |
| RVP (kPa)           |                                 | 62.0  |         | 69.0  |

Source: US Grains Council

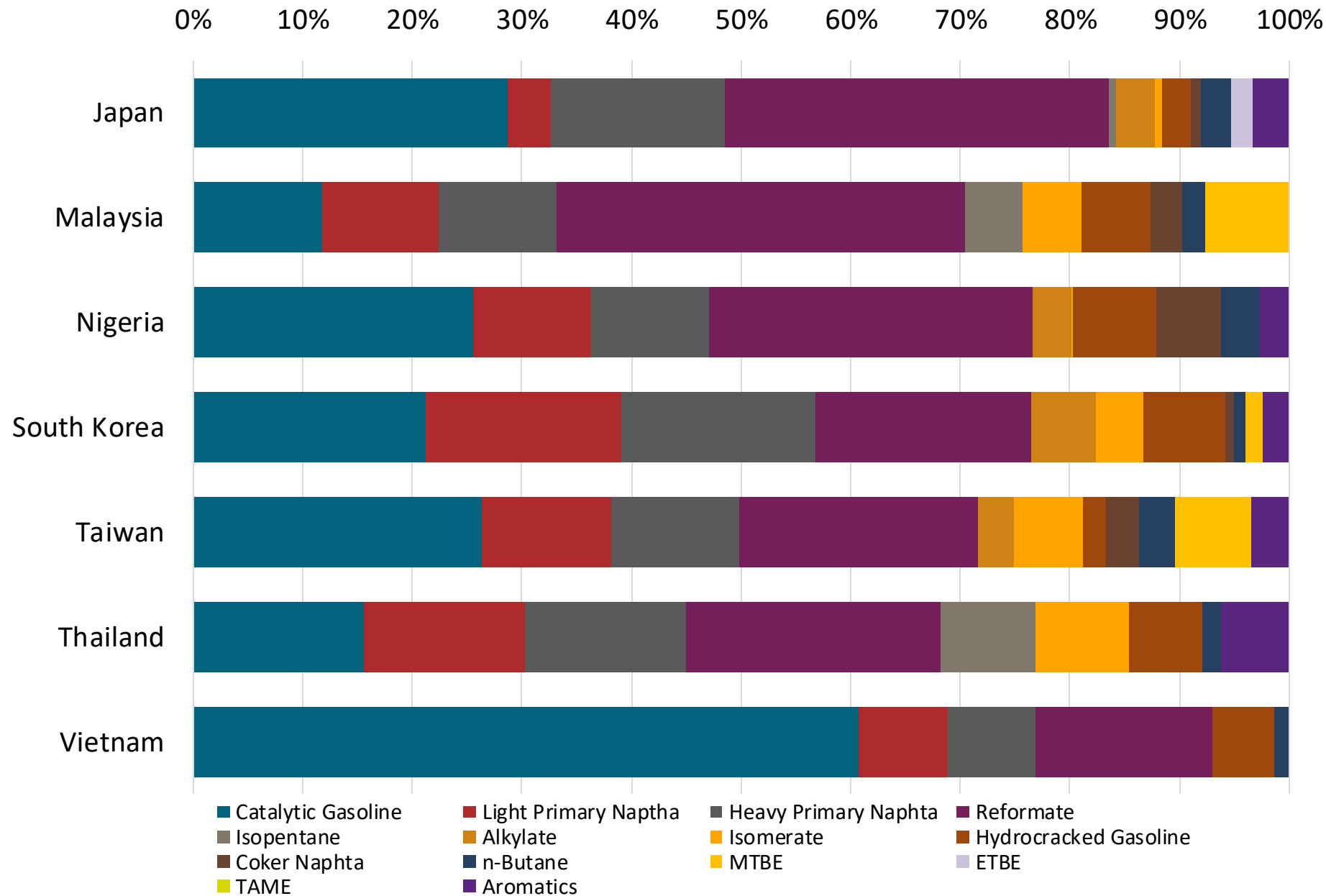


# Gasoline Component Blending in Asia Africa

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Asia Africa are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics

Source: Faro90



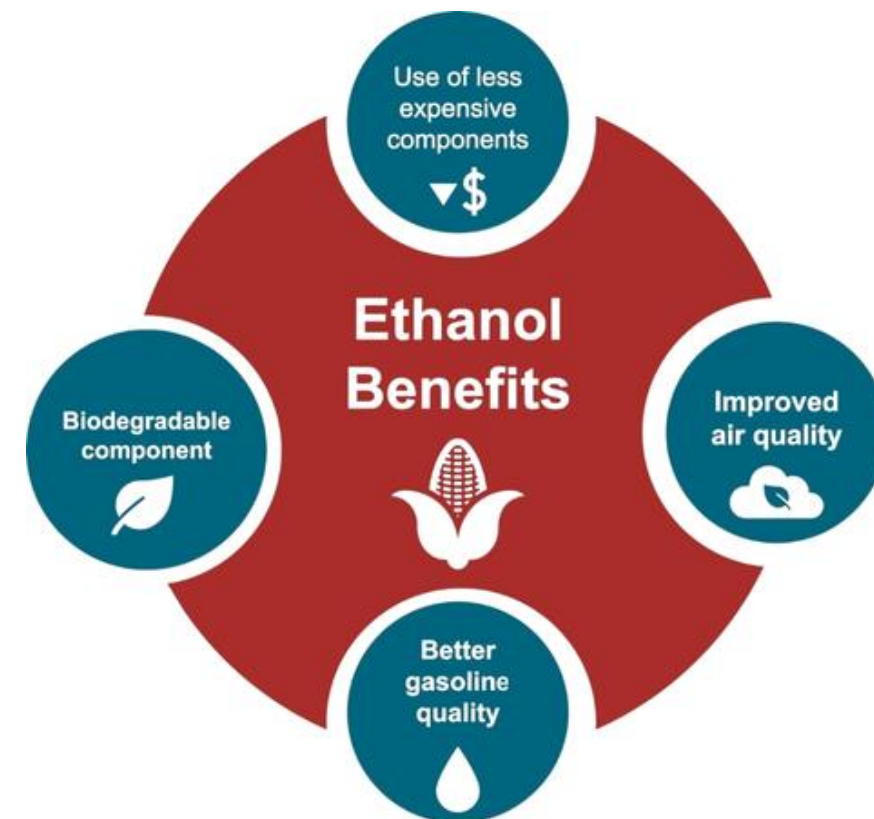
# Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

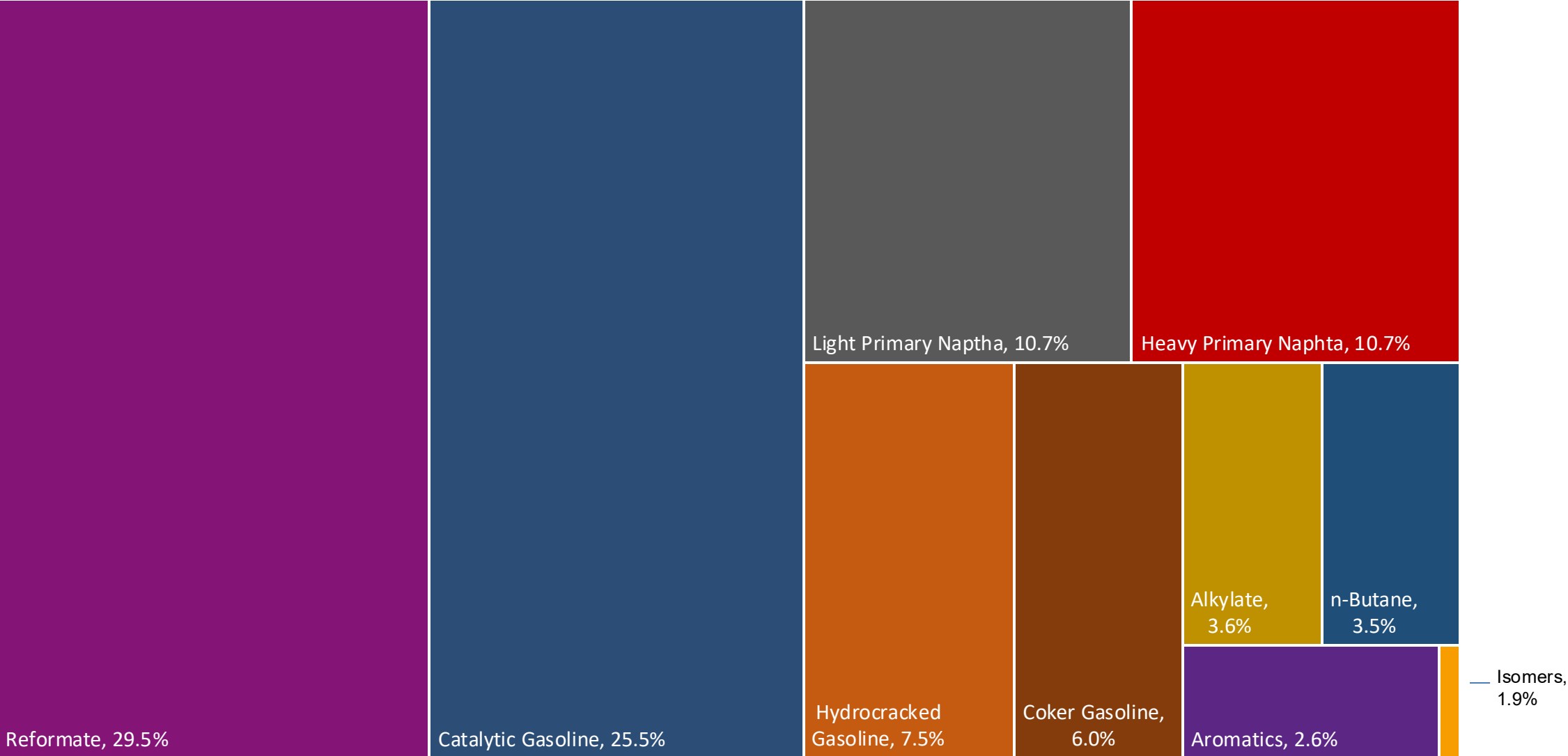
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

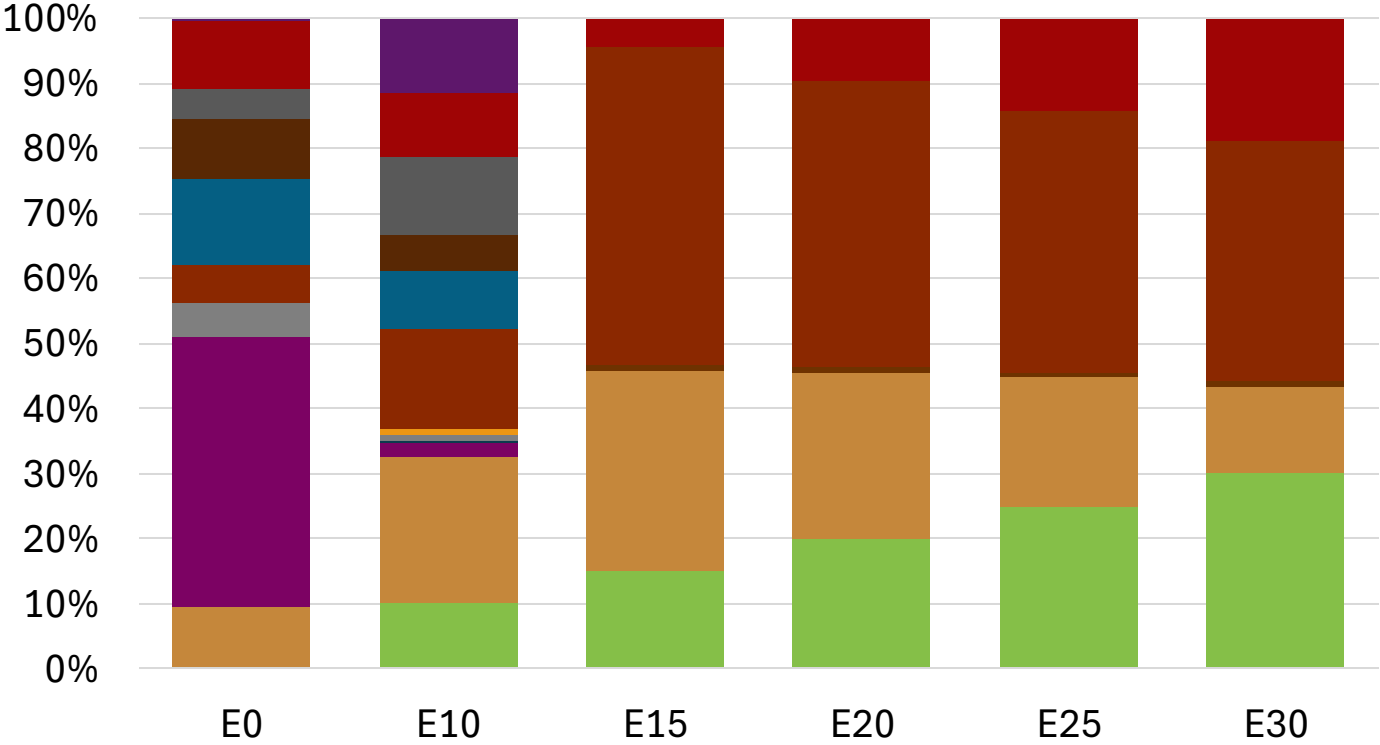
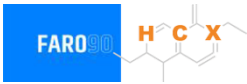


# Nigeria Available Blending Components



Source: Faro90

# Nigeria – Gasoline 91 – Constant Octane

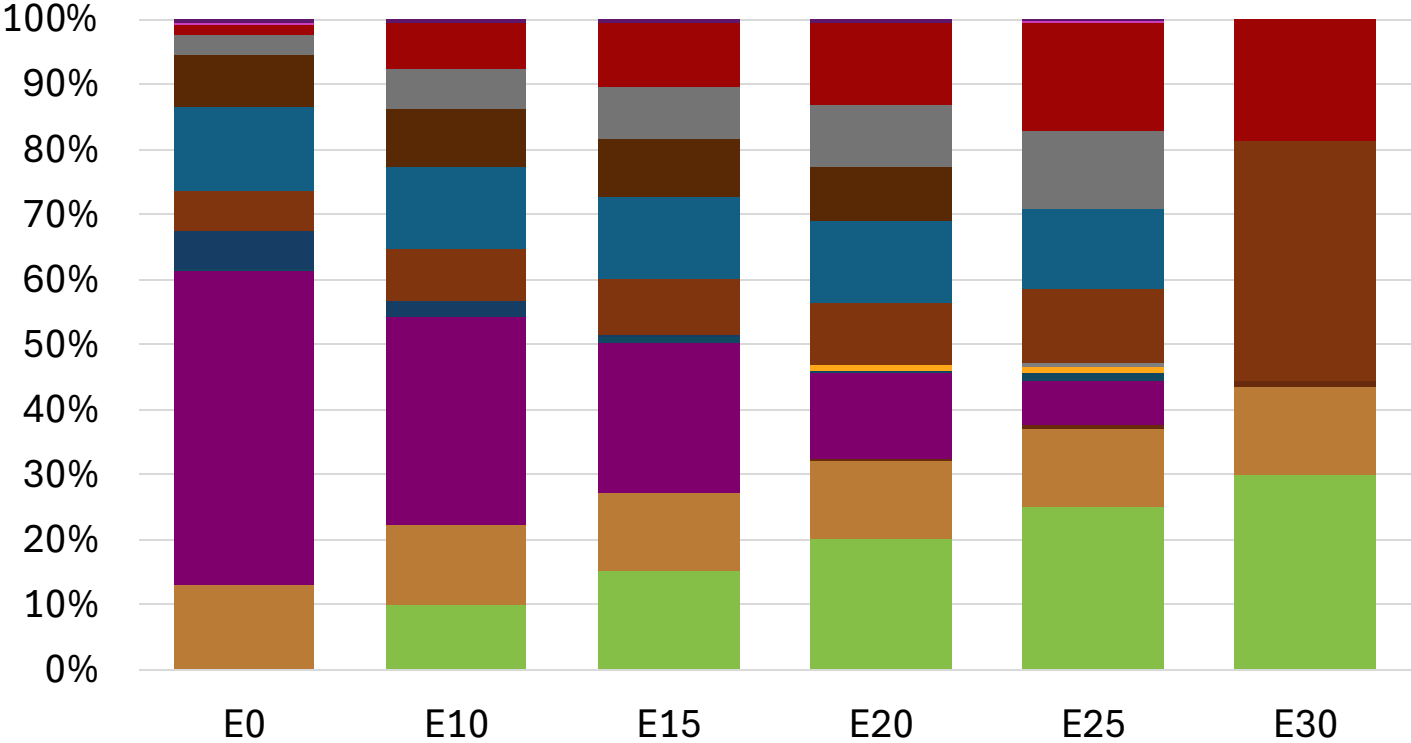


Average CIF 2024  
Prices

|                     |          |          |          |          |          |          |
|---------------------|----------|----------|----------|----------|----------|----------|
| Octane<br>(RON)     | 91.0     | 91.0     | 91.0     | 92.5     | 93.9     | 95.2     |
| Price<br>(US\$/gal) | \$ 2.230 | \$ 2.194 | \$ 2.131 | \$ 2.090 | \$ 2.048 | \$ 2.005 |

Source: Faro90

# Nigeria - Gasoline 95 - Constant Octane

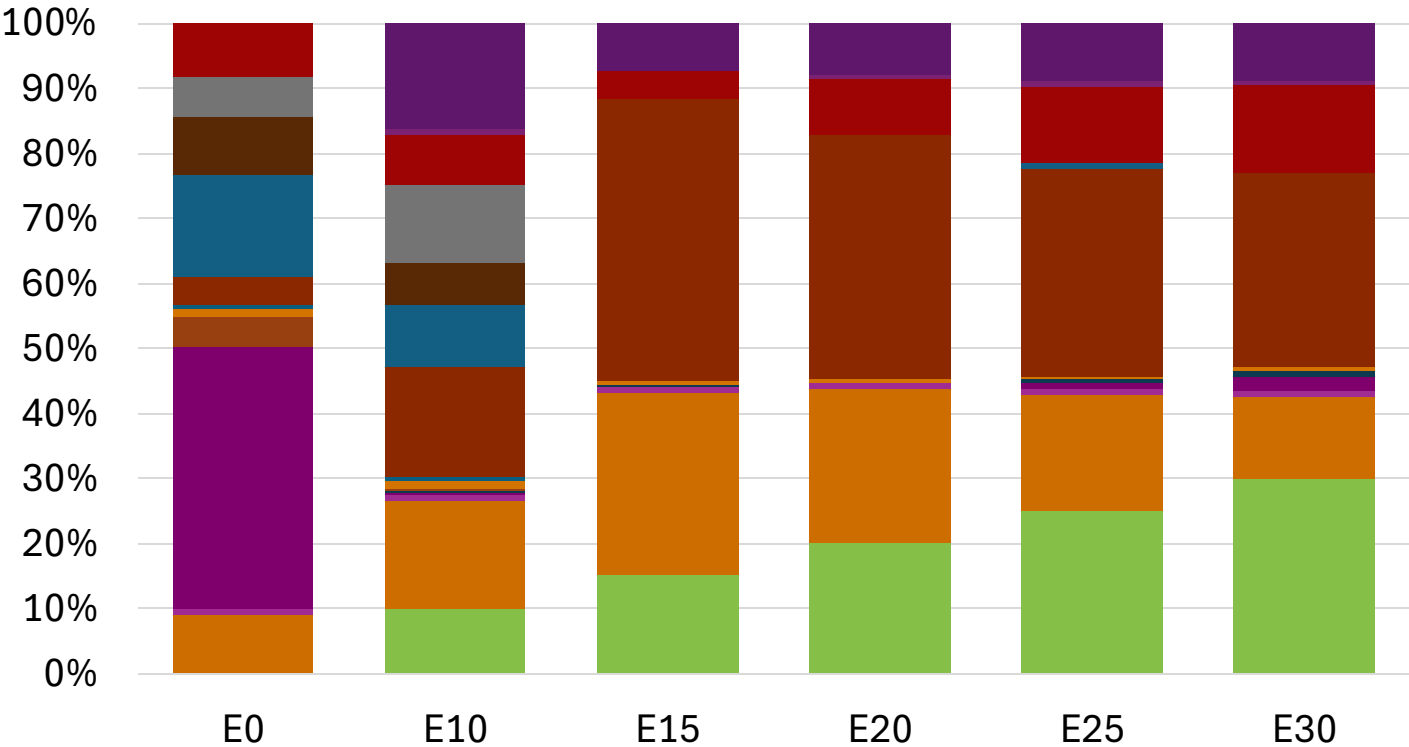


Average CIF 2024 Prices

|                  |          |          |          |          |          |          |
|------------------|----------|----------|----------|----------|----------|----------|
| Octane (RON)     | 95.0     | 95.0     | 95.0     | 95.0     | 95.0     | 95.2     |
| Price (US\$/gal) | \$ 2.416 | \$ 2.293 | \$ 2.220 | \$ 2.142 | \$ 2.057 | \$ 2.005 |

Source: Faro90

# Nigeria – Gasoline 91 – Increased Octane

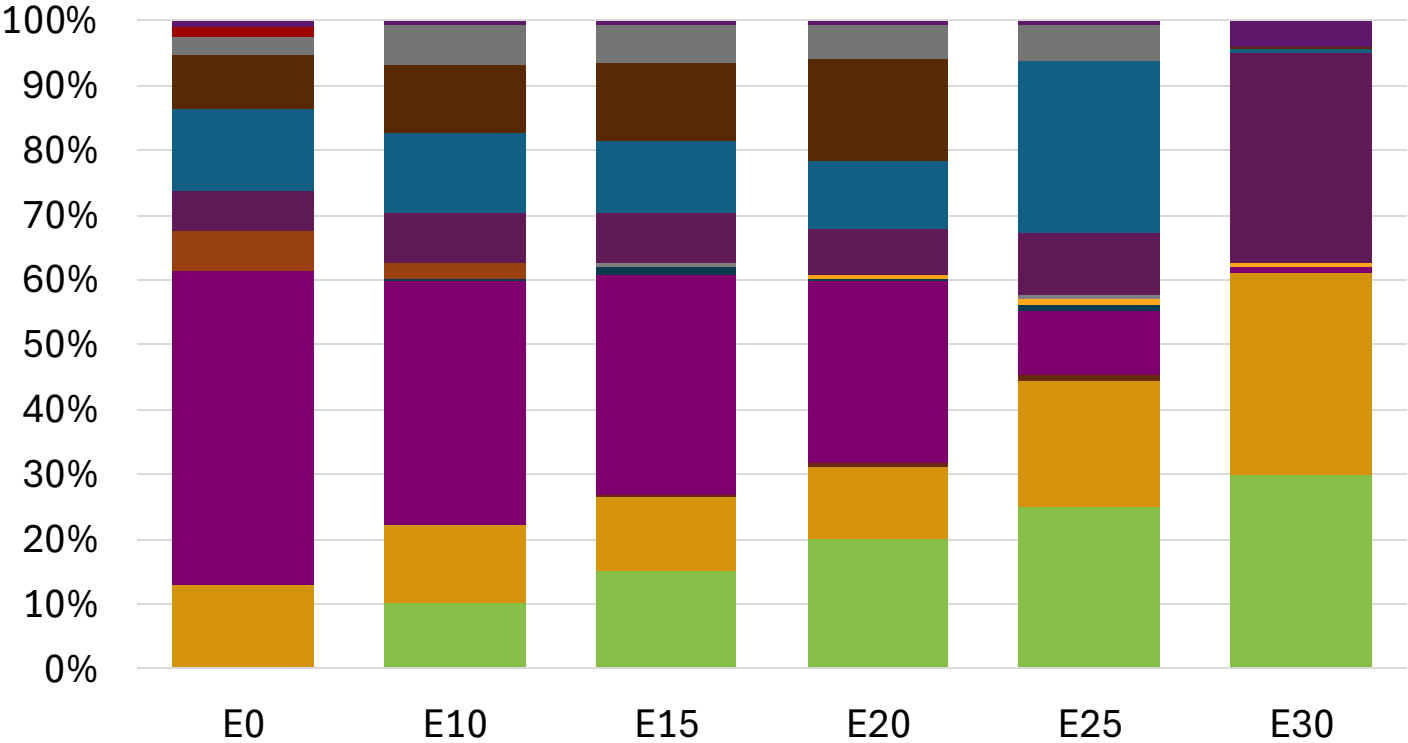


Average CIF 2024  
Prices

|                  |          |          |          |          |          |          |
|------------------|----------|----------|----------|----------|----------|----------|
| Octane (RON)     | 91.0     | 91.6     | 92.8     | 94.6     | 96.8     | 98.6     |
| Price (US\$/gal) | \$ 2.230 | \$ 2.230 | \$ 2.230 | \$ 2.230 | \$ 2.230 | \$ 2.230 |

Source: Faro90

# Nigeria – Gasoline 95 – Increased Octane



Average CIF 2024  
Prices

# Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

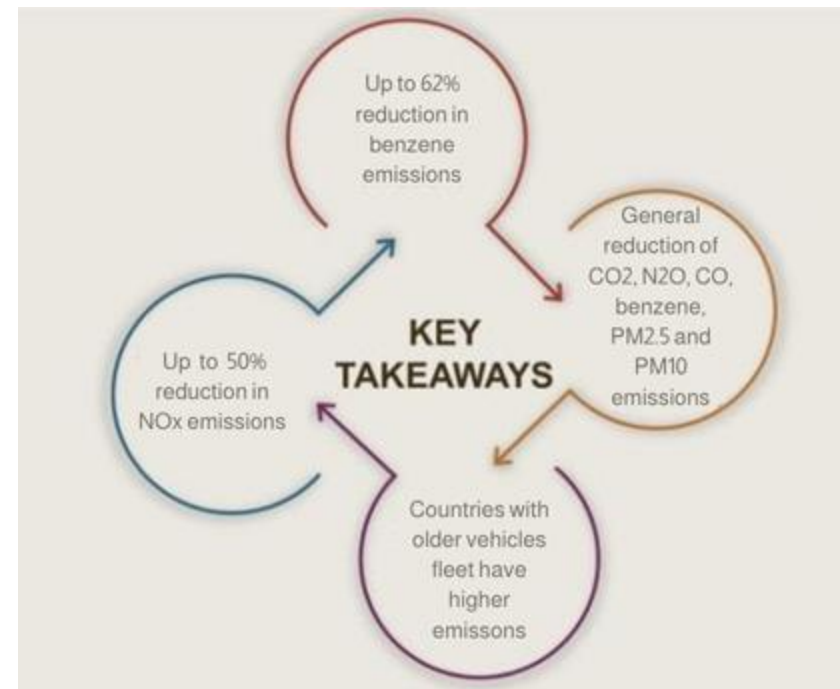
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet\*.

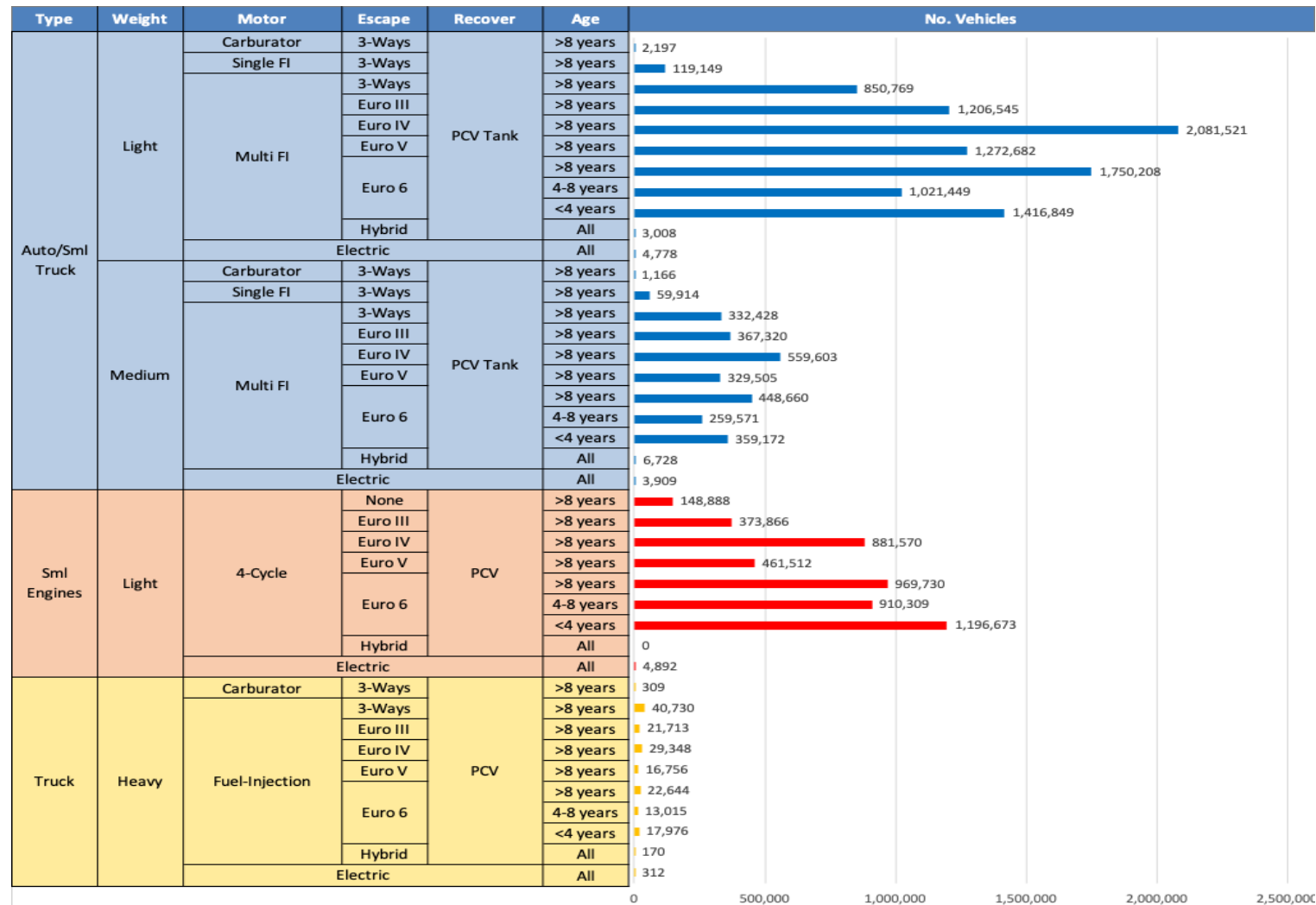
\*Source: Bureau of transportation statistics.

## Main Results





# Gasoline Vehicle Fleet - Nigeria

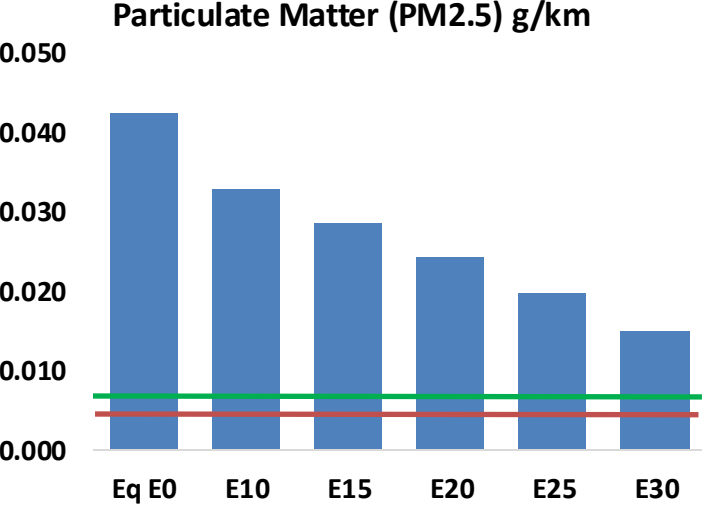
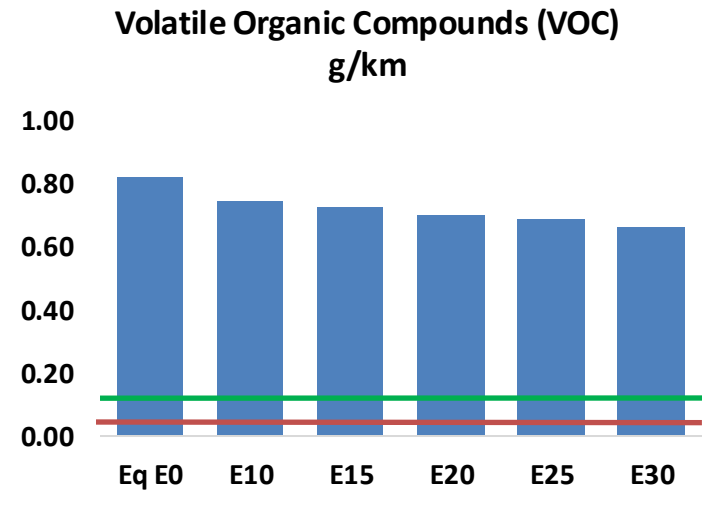
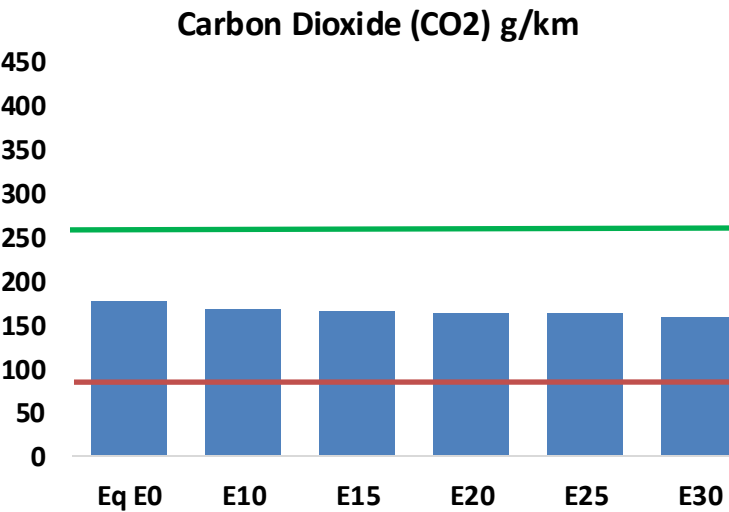
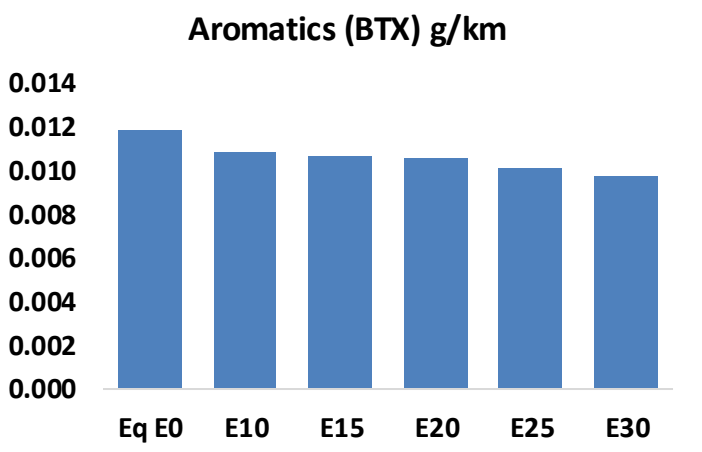
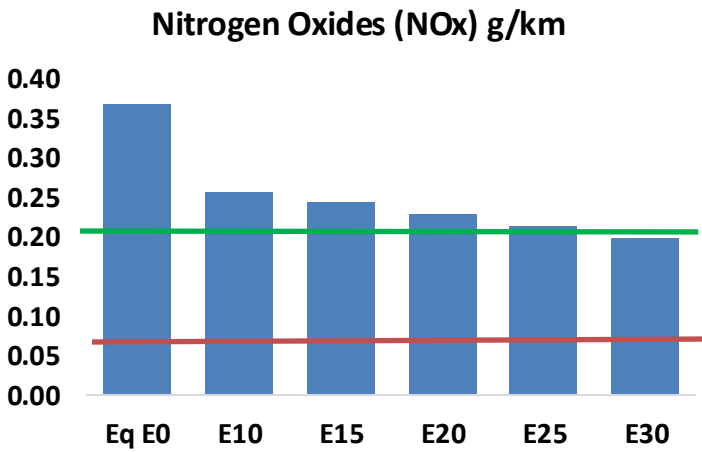
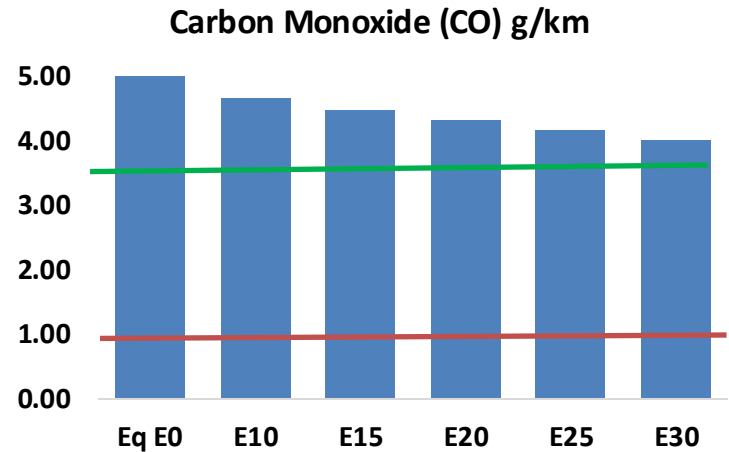
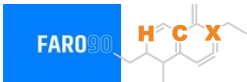


Vehicle Fleet: 17,567,546  
Gasoline Vehicle Fleet: 16,203,007

Source: Nigeria Stat

# Nigeria – Vehicular Emissions

TIER III BIN 125 United States  
Euro 6 Standard



# Nigeria – Vehicular Emissions

| Vehicular Fleet Emissions (kg/day) | Eq E0      | E5         | E10        | E15        | E20        | E25        | E30        | Reduction (%) |      |
|------------------------------------|------------|------------|------------|------------|------------|------------|------------|---------------|------|
|                                    |            |            |            |            |            |            |            | E10           | E20  |
| CO                                 | 2,051,399  | 1,948,219  | 1,845,039  | 1,771,126  | 1,701,643  | 1,649,693  | 1,581,391  | -10%          | -17% |
| CO2                                | 70,013,767 | 68,213,413 | 66,513,078 | 65,175,815 | 64,514,085 | 64,188,202 | 63,005,028 | -5%           | -8%  |
| VOC                                | 323,008    | 308,955    | 294,902    | 285,489    | 276,985    | 270,738    | 261,371    | -9%           | -14% |
| NOx                                | 145,686    | 109,264    | 101,980    | 96,116     | 90,646     | 84,614     | 78,233     | -30%          | -38% |
| SOx                                | 3,421      | 3,163      | 2,904      | 2,686      | 2,476      | 2,248      | 2,009      | -15%          | -28% |
| NH3                                | 28,328     | 28,047     | 27,767     | 27,899     | 28,212     | 28,433     | 28,617     | -2%           | 0%   |
| N2O                                | 1,245      | 1,238      | 1,233      | 1,239      | 1,265      | 1,279      | 1,293      | -1%           | 2%   |
| CH4                                | 70,376     | 70,376     | 70,376     | 71,432     | 72,980     | 74,317     | 75,584     | 0%            | 4%   |
| Lead                               | -          | -          | -          | -          | -          | -          | -          | 0%            | 0%   |
| Butadiene                          | 2,053      | 1,985      | 1,917      | 1,879      | 1,849      | 1,828      | 1,792      | -7%           | -10% |
| Acetaldehyde                       | 5,652      | 6,487      | 9,518      | 14,494     | 19,747     | 22,564     | 26,662     | 68%           | 249% |
| Formaldehyde                       | 21,184     | 20,595     | 24,008     | 28,191     | 29,534     | 32,672     | 35,568     | 13%           | 39%  |
| Benzene                            | 4,692      | 4,509      | 4,272      | 4,205      | 4,170      | 3,988      | 3,859      | -9%           | -11% |
| PM2.5                              | 4,605      | 4,089      | 3,573      | 3,103      | 2,635      | 2,138      | 1,620      | -22%          | -43% |
| PM10                               | 12,129     | 10,770     | 9,411      | 8,172      | 6,940      | 5,631      | 4,267      | -22%          | -43% |

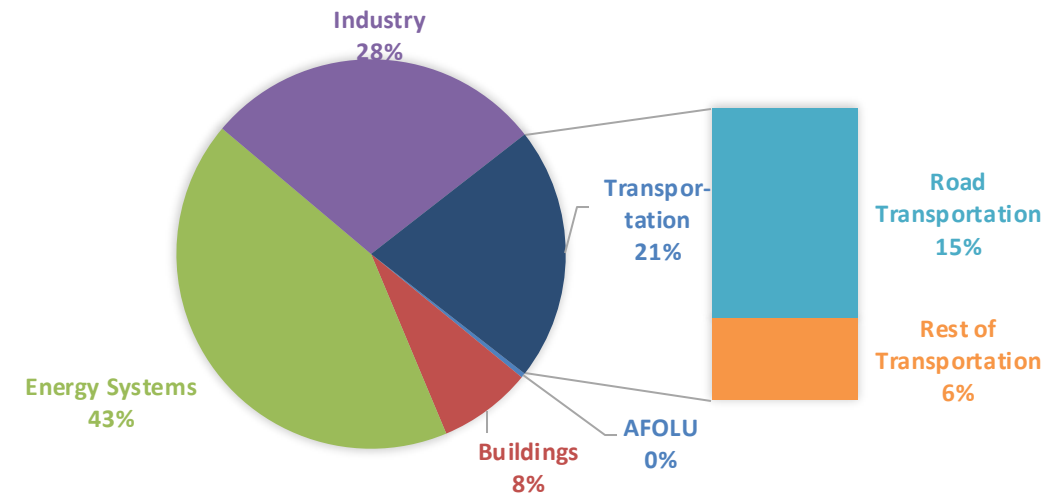
# Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

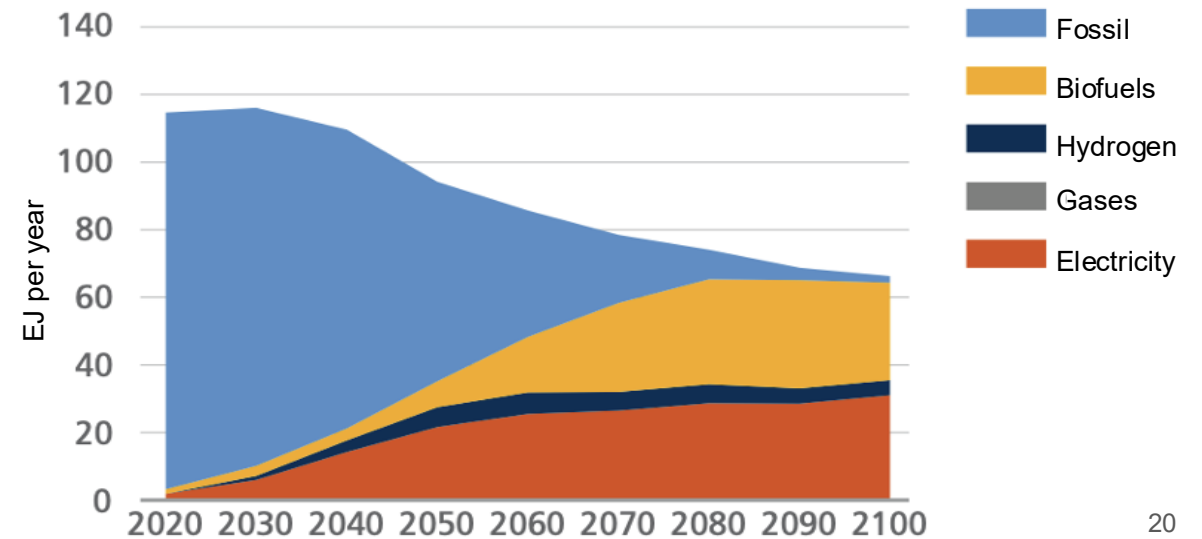
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

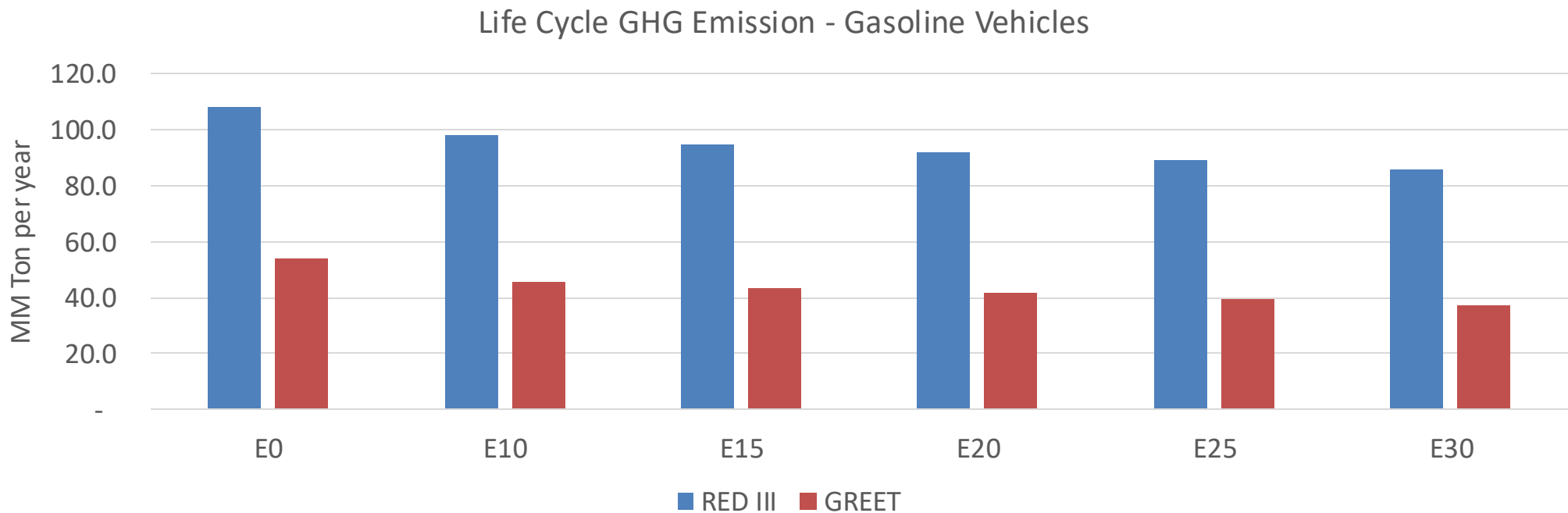
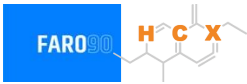
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



# Nigeria – Gasoline Vehicle Fleet Life Cycle GHG Emissions



|                                  |       |
|----------------------------------|-------|
| Country Emissions 2024 (MMT/año) | 413.6 |
| Target @ 2035 (MMT/year)         | 267.2 |
| Delta                            | 146.3 |

| %Reduction vs E0 | E0   | E10   | E15   | E20   | E25   | E30   |
|------------------|------|-------|-------|-------|-------|-------|
| GREET 2024       | 0.0% | 15.0% | 19.1% | 22.8% | 26.4% | 30.7% |
| RED II/III       | 0.0% | 9.3%  | 12.3% | 14.9% | 17.5% | 20.4% |

| % Target@2035 | E0   | E10  | E15  | E20   | E25   | E30   |
|---------------|------|------|------|-------|-------|-------|
| GREET 2024    | 0.0% | 5.6% | 7.1% | 8.4%  | 9.7%  | 11.3% |
| RED II/III    | 0.0% | 6.9% | 9.0% | 11.0% | 12.9% | 15.0% |

Source: Greet, RED II/III, climateactiontracker.org, F90